



UNIBUS LIVE

Real-Time University Bus Tracking — Live Map, Arrival Times & Alerts, in English & Bangla

TEAM UNIBUS

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INTRODUCTION

UniBus Live shows every university bus **live on a map**, with the **arrival time for each stop**. A GPS tracker fixed inside the bus reports its position every 5 seconds; our server cleans the data, calculates arrival times, and pushes updates to every student's phone **within seconds**.

OBJECTIVES

- Let riders see the bus and their stop's arrival time before leaving home
- Track without depending on the driver — the fixed tracker works by itself
- Keep the whole system affordable on a single low-cost cloud server
- Serve every student in both **English and Bangla**

PROBLEM STATEMENT

Students wait at stops with no idea whether the bus left five minutes ago or is fifteen minutes away — causing missed buses, long waits and crowding. Commercial trackers are costly, operator-facing and English-only.

KEY HARDWARE & STACK

Concox GT06S GPS tracker
fixed in the bus · GPS + GSM · fully automatic

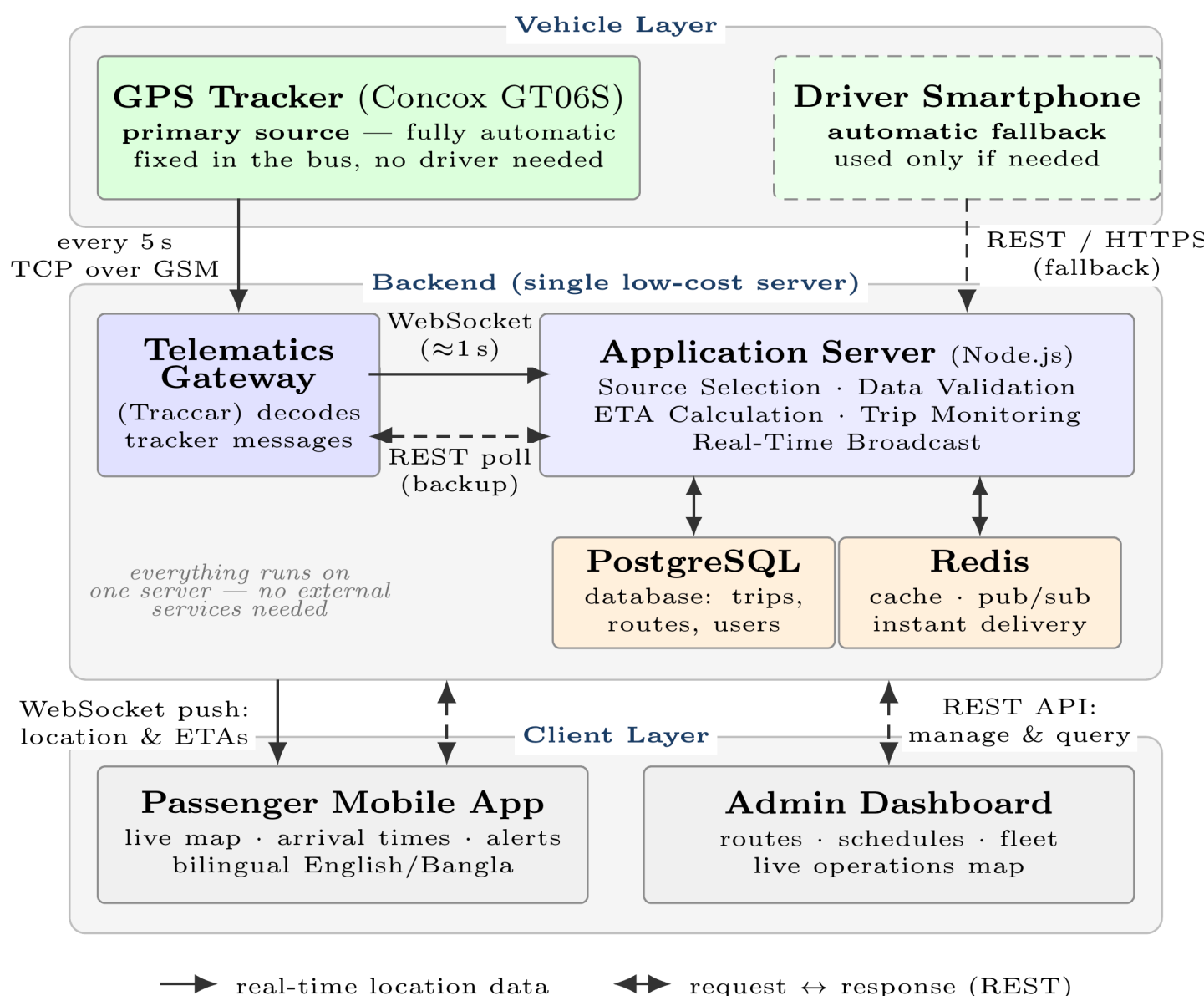
Driver smartphone
backup source only, used if the tracker is silent

Cloud server (VPS)
one low-cost machine runs everything

Passenger smartphones
live map + notifications for every rider

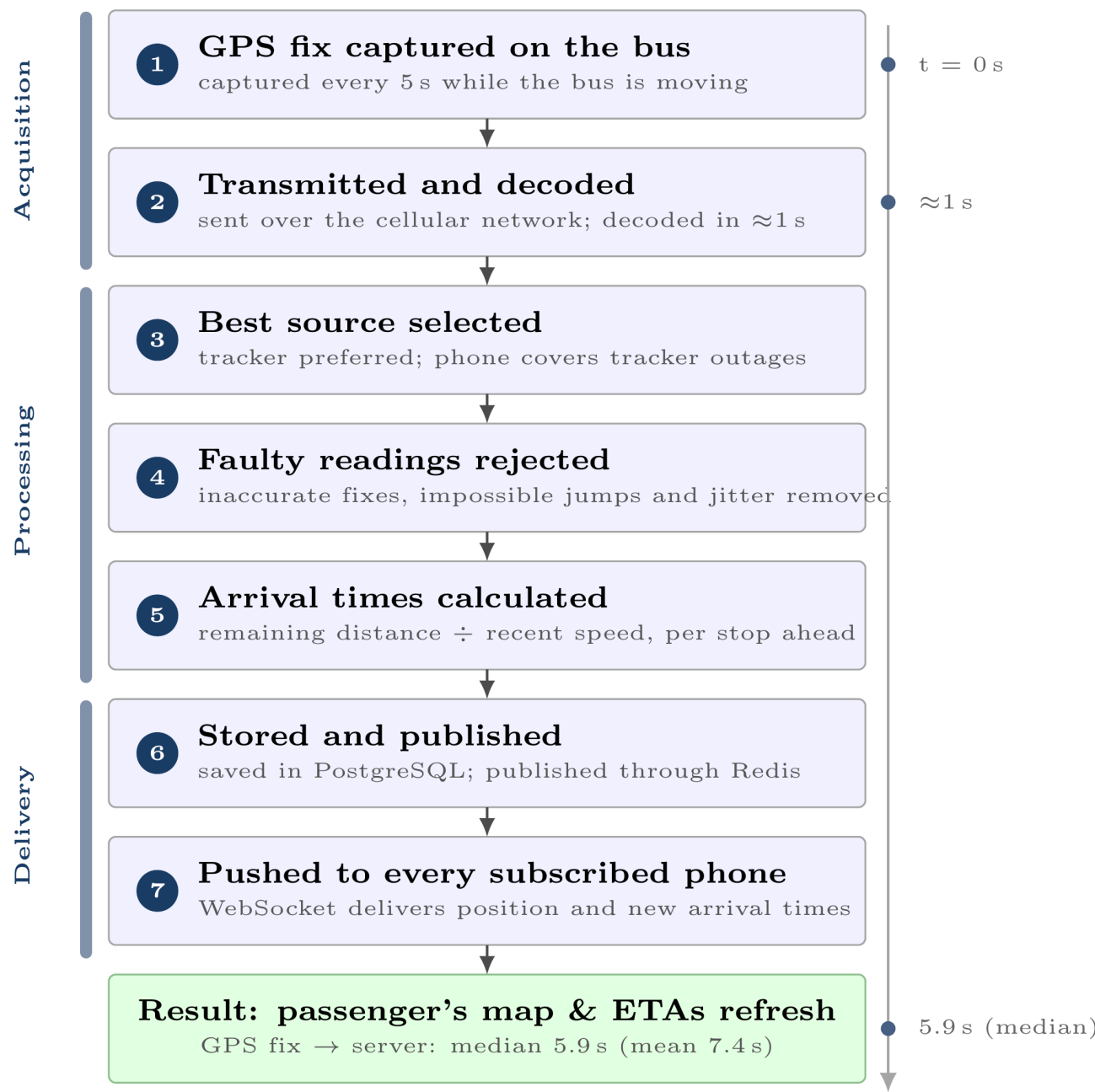
Node.js PostgreSQL Redis Traccar Socket.IO React Native
Next.js

SYSTEM ARCHITECTURE & SETUP



Three layers: the bus, one cloud server, and the users — the tracker needs no driver action at all.

METHODOLOGY



The journey of one position report — from the GPS fix on the bus to the rider's screen.

RESULTS & OUTPUT

5.9 s
median delay,
GPS fix → server

46,548
GPS reports
handled

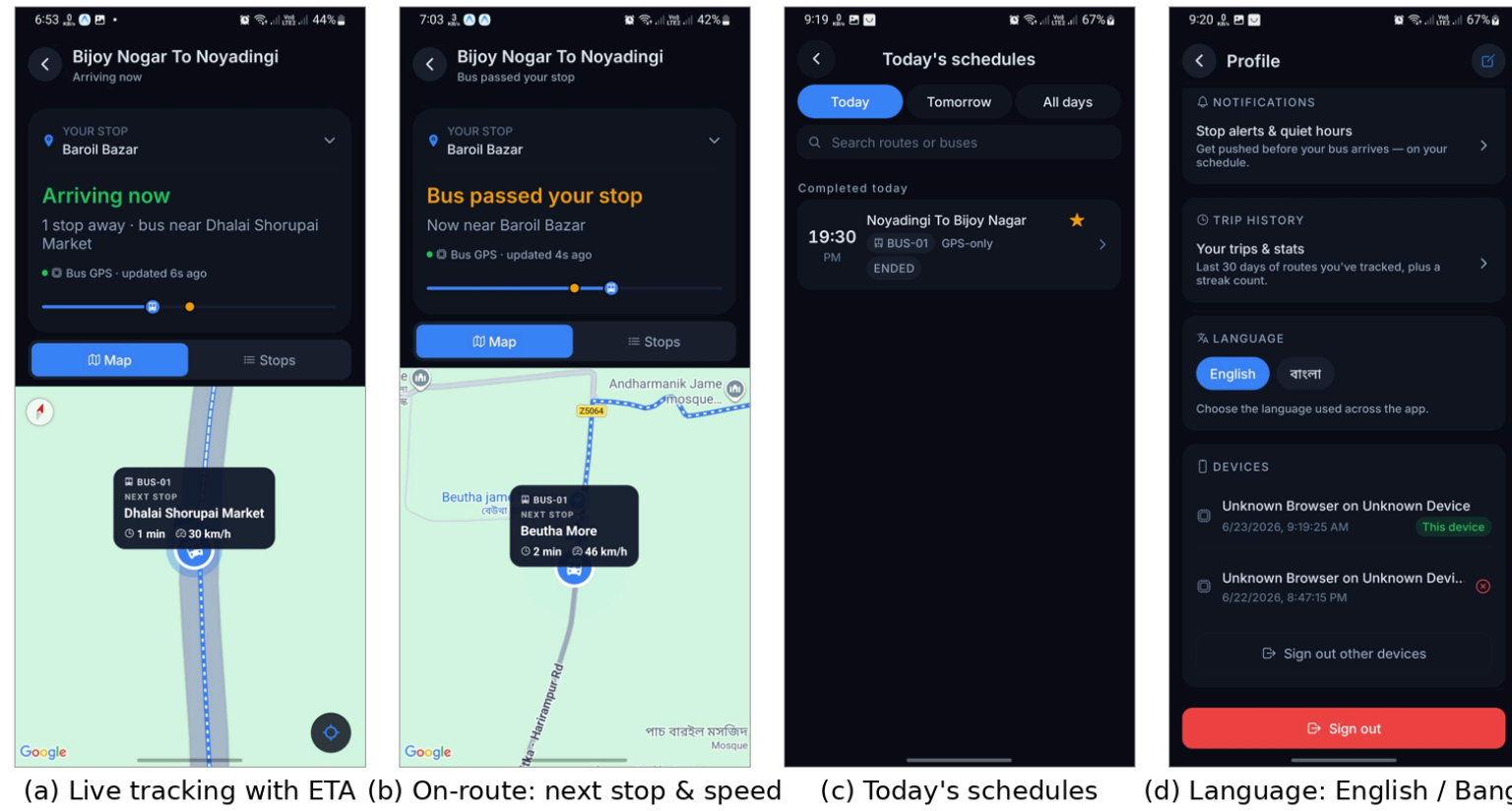
71
real trips tracked
across 9 routes

64 / 36
% trips by tracker
vs phone backup

80%
arrival alerts
read by riders

60×
less idle reporting
after our fix

Proven in a 7-week pilot on the real university fleet — 3 buses · 9 routes · 38 riders · zero simulation



The bilingual passenger app: live tracking with ETA · on-route status · today's schedules · English/Bangla toggle

FUTURE SCOPE

- Smarter ETAs** — an AI model trained on our logged trip history
- Trackers on every bus**, plus seat-occupancy (crowd) info
- Automatic on-time reports** for the university administration

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